

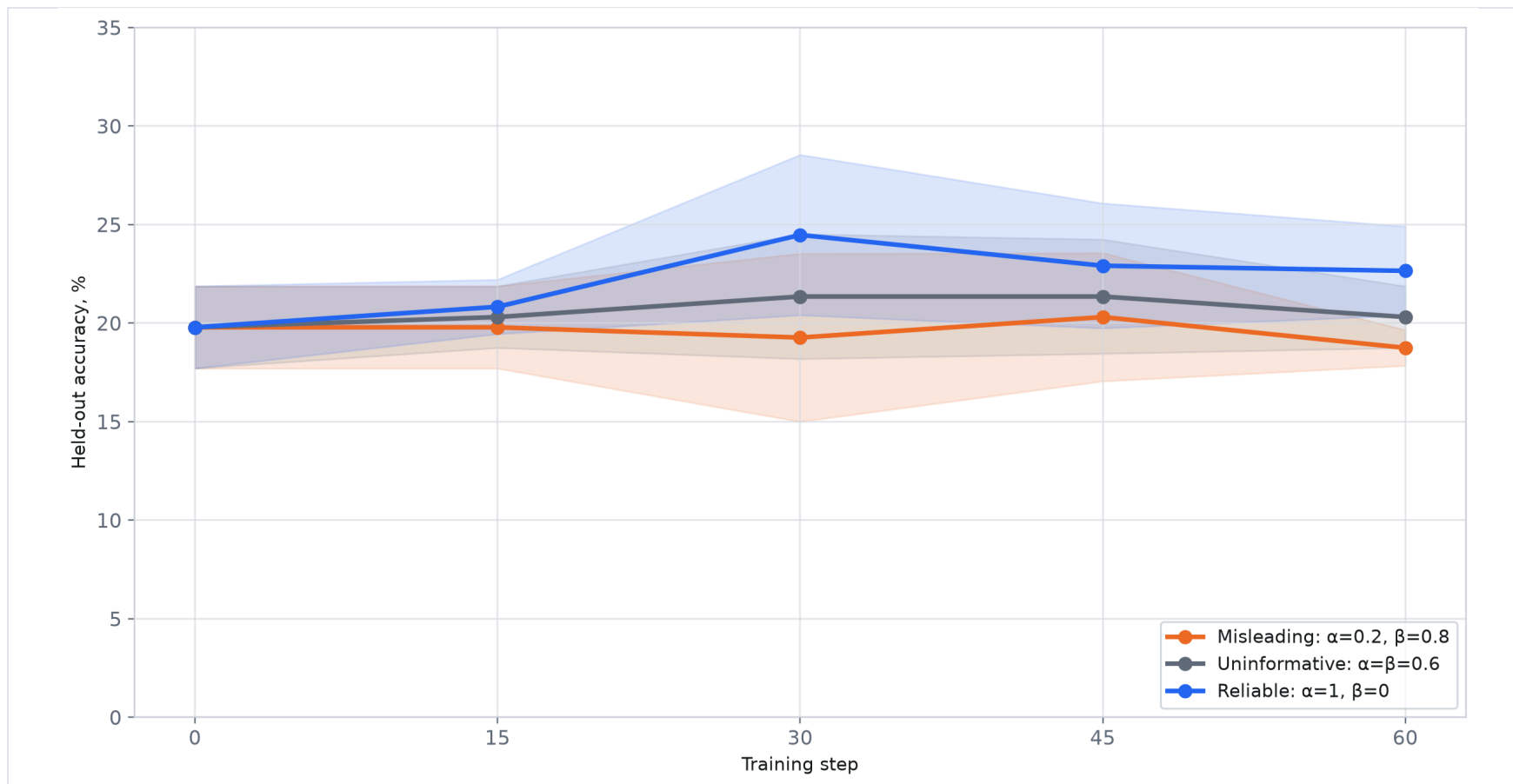
A reliable verifier gives the **best final result**

QWEN3-0.6B-BASE + LORA

3 REGIMES × 3 SEEDS

60 STEPS / G=8

10 H 05 MIN

**22.7%**

Reliable verifier. Accuracy peaked at step 30 at 24.5%; gains did not continue.

20.3%

The uninformative verifier finished close to the 19.8% baseline.

18.8%

The misleading verifier finished below baseline.

Lines show means across three seeds; bands show ± 1 SE. The two step-60 evaluations are averaged.

The three regimes follow the **expected order**

misleading verifier

-1.0 pp

SE 2.3 pp

uninformative verifier

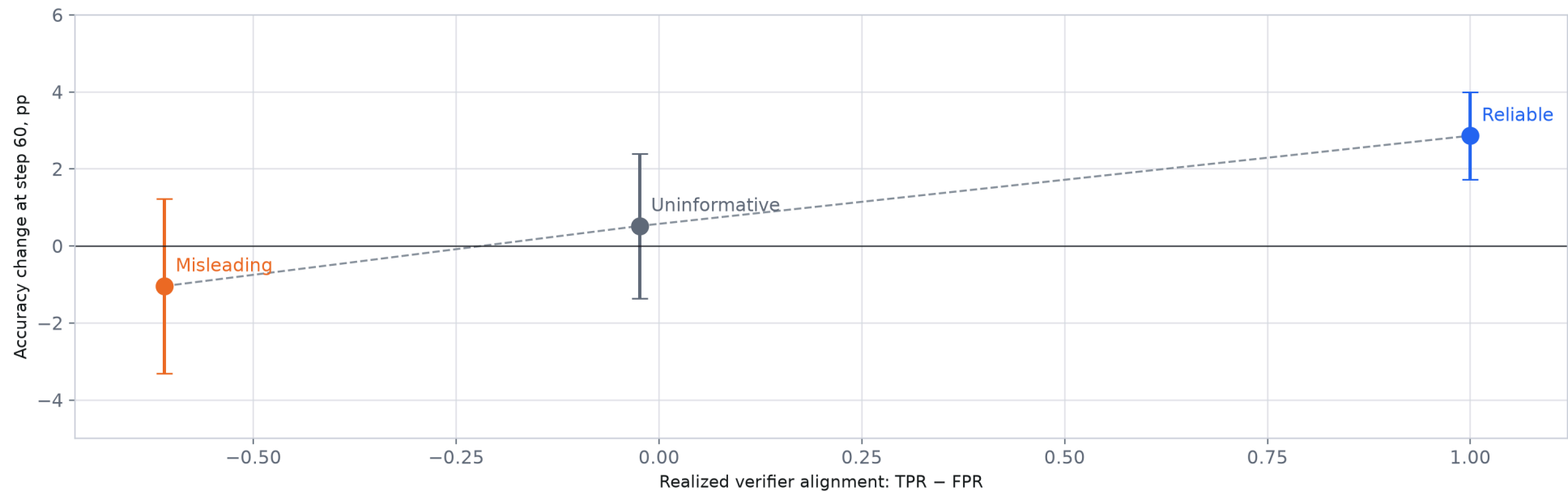
+0.5 pp

SE 1.9 pp

reliable verifier

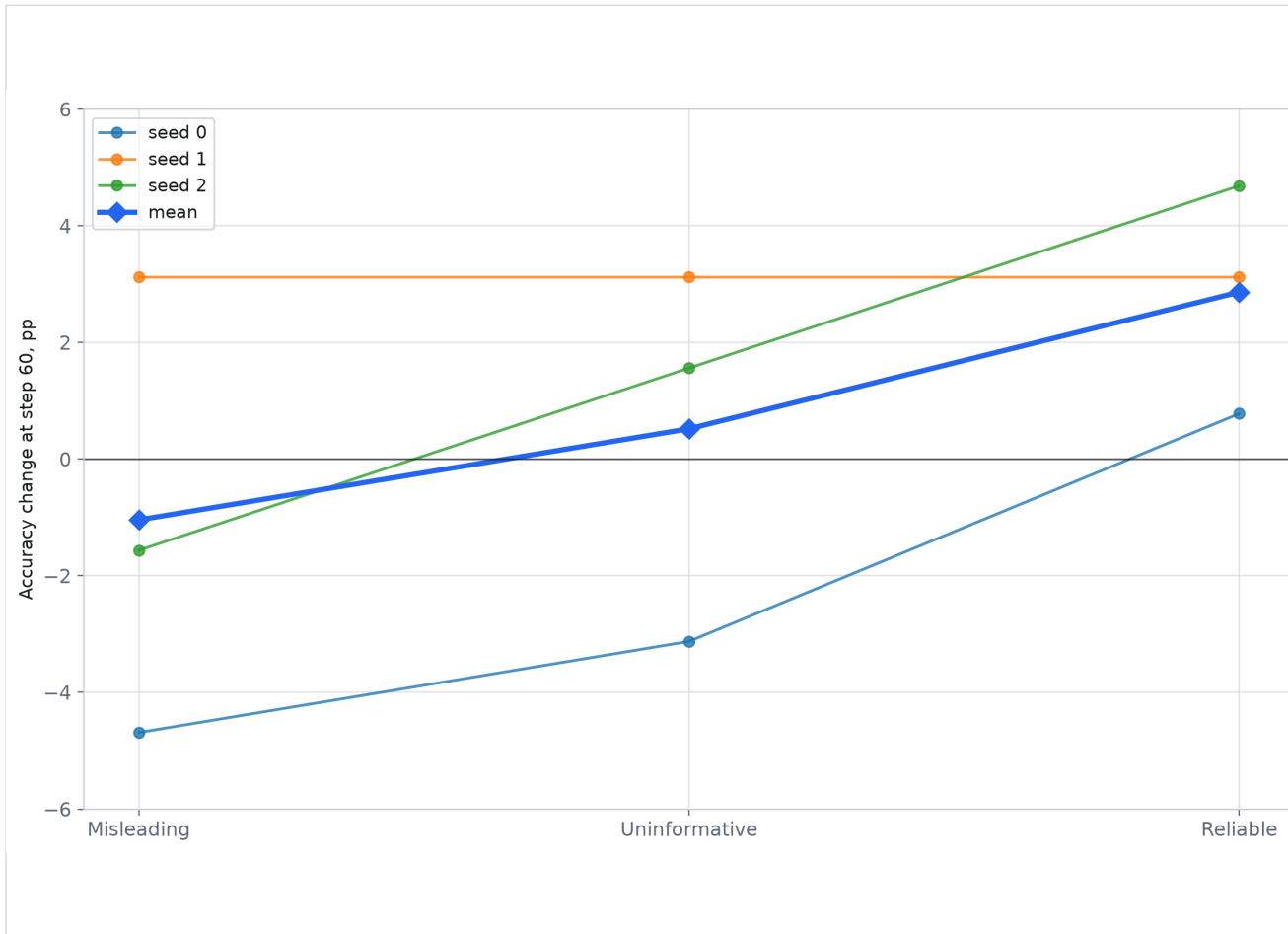
+2.9 pp

SE 1.1 pp



Reliable-versus-misleading gap: 3.9 pp. Realized TPR - FPR was 1.000, -0.024, and -0.610.

The expected order holds for **two of three seeds**



Seed 0

Reliable beats uninformative; uninformative beats misleading.

Seed 1

All three regimes gained the same +3.1 pp. No separation.

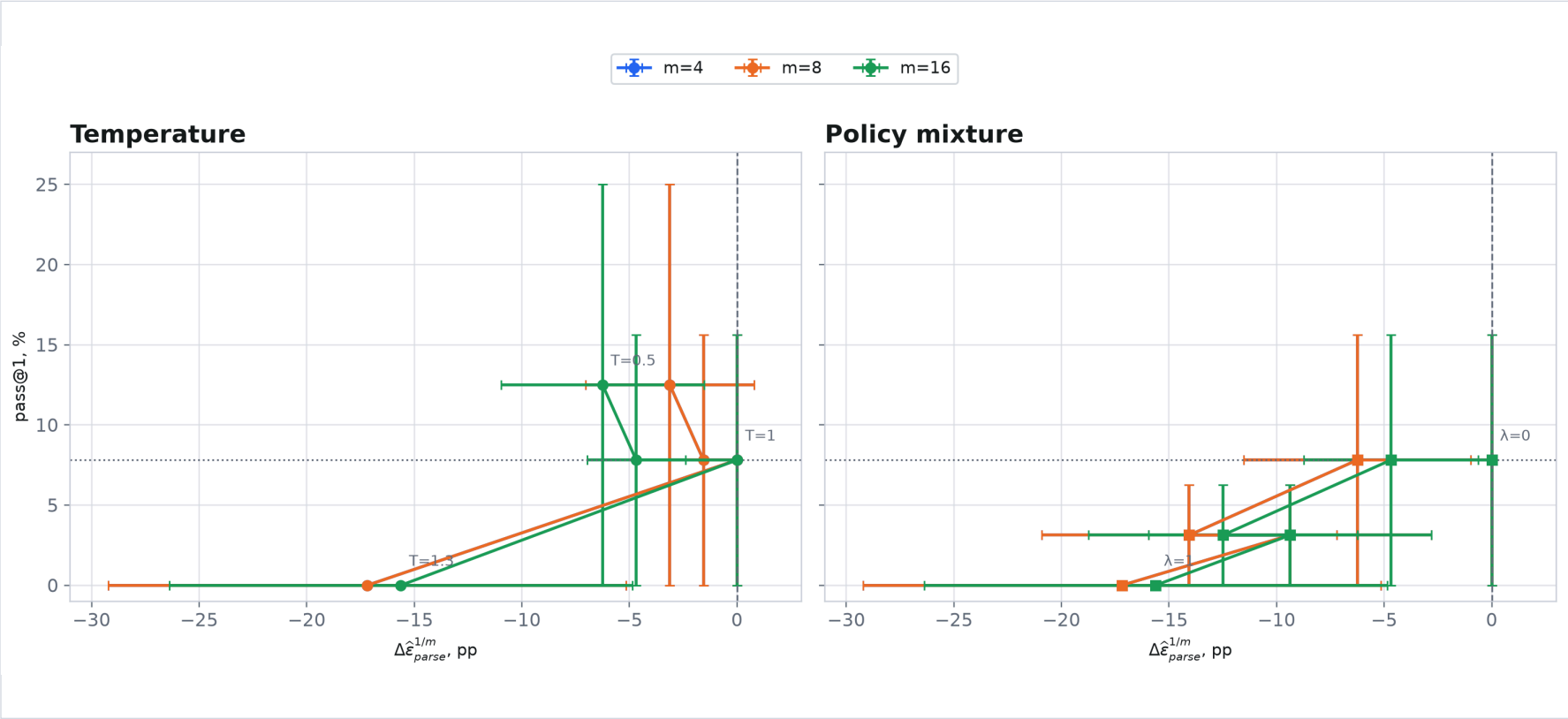
Seed 2

The expected order returns. Reliable gains 4.7 pp; misleading loses 1.6 pp.

The mean has the expected order, but $n=3$ is too small for a firm statistical conclusion.

Interventions did not expand useful support

QWEN3-0.6B-BASE Π : T=1.0 8 PROBLEMS $\times$ 1 SEED M=32 / K=8 M=4, 8, 16



$\Delta \epsilon_{parse} \leq 0$

Neither temperature nor policy mixture produced more novel parseable mass than baseline.

$\epsilon_{correct} = 0$

No condition reached a new correct answer class outside baseline support.

$\Delta \epsilon_{parse}$ ranged from -17.2 to 0 pp. Error bars are ± 1 problem-level SE. This is answer-level novelty; one seed is insufficient for a stable statistical conclusion.

Evidence supports the hypothesis; **final validation is still pending**

What we observe

- | Mean performance rises with verifier quality.
- | Reliable and misleading regimes differ by 3.9 pp at the final step.
- | Realized TPR and FPR confirm that the verifiers behaved as intended.

Limits

- | Only three seeds. The 95% confidence interval for gain includes zero in every regime.
- | No regime plateaued across all three seeds; neither the 90% nor 95% threshold was reached.
- | The base-solved metric contains only 3-4 problems per seed, too few to assess forgetting.
- | The exploration pilot found $\Delta\epsilon_{\text{parse}} \leq 0$ and $\epsilon_{\text{correct}} = 0$ on eight problems and one seed.

Stop new local training runs. If stronger evidence is needed, first reevaluate the nine saved adapters on a larger fixed set. Reserve the full H2 protocol for GPUs: 25 regimes × 3 seeds, at least 600 steps.